

IT549: Deep Learning –

Lab:2 Assignment

GloVe Pretrained Embeddings for Movie Text Prediction

Dataset: Movie Dataset (Kaggle) - <https://www.kaggle.com/datasets/figolm10/movie-dataset>

Use only columns: genre, keywords, tagline, overview, voting_average

Submission: Public GitHub repository link only

Objective

Demonstrate the use of pretrained GloVe word embeddings for two predictive tasks using movie metadata text: (1) regression to predict voting_average from a single text column, and (2) multi-label classification to predict movie genres from a single text column. You will also conduct text-based analysis to interpret which words are most frequent and most indicative of each genre.

Dataset Scope

You must use only the following fields:

- overview (text)
- tagline (text)
- keywords (text)
- genre (multi-label target: a movie may belong to multiple genres)
- voting_average (regression target)

Train separate models where the input is one text column at a time. Do not concatenate multiple columns for the primary experiments.

Tasks

Task 1 - Data Preparation

- Load the dataset and retain only the allowed columns.
- Perform text preprocessing: lowercase; remove URLs, punctuation, and numbers; tokenize; optional lemmatization.
- Create reproducible train/validation/test splits (suggestion: 70/15/15).

Task 2 - GloVe Embedding Pipeline

- Download and load pretrained GloVe vectors (100D or 200D; clearly state which dimension you use).
- Report embedding coverage: percentage of unique dataset tokens present in GloVe.
- Construct document embeddings using TF-IDF weighted averaging of GloVe word vectors.
- Keep embedding dimensionality consistent across all experiments.

Task 3 - Model A: Rating Prediction (Regression)

- Select one text column as input (overview OR tagline OR keywords).
- Train a neural regression model to predict `voting_average` from the document embedding.
- Report MSE and RMSE on the test set.
- Include a baseline model that predicts the global mean rating.
- Repeat for at least two different single-text inputs and compare performance.

Task 4 - Model B: Genre Prediction (Multi-Label Classification)

- Select one text column as input (overview OR tagline OR keywords).
- Train a multi-label classifier to predict genres.
- Use sigmoid outputs and an appropriate loss such as `BCEWithLogitsLoss`.
- Report micro-F1 and macro-F1; additionally report Hamming loss or Jaccard score.
- Repeat for at least two different single-text inputs and compare results.

Task 5 - Frequent Words per Genre

- For each genre, compute the top 10 most frequent content words after preprocessing.
- For each genre, compute the bottom 10 least frequent words (apply a minimum frequency threshold such as ≥ 3).
- Present results clearly in tables and briefly interpret patterns.

Task 6 - Genre-Indicative Words Using TF-IDF

- Use TF-IDF combined with a linear model (e.g., logistic regression per genre) to identify indicative words.
- Extract the highest positive-weight words for each genre.
- For each genre, report 10 indicative words and provide a short interpretation explaining why they suggest that genre.

Deliverables (GitHub Only)

Your public GitHub repository must include:

- A README containing assignment title, your name, and your ID along with the project details.
- Complete runnable code covering preprocessing, embeddings, training, evaluation, and analysis.
- A results summary comparing performance across different input columns.